

BLACK FRIDAY SALES

Exploratory Data Analysis — Project Report

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Dataset: Black_Friday_Sales.csv

Tool: Python (Pandas, Matplotlib, Seaborn) — Google Colab

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1. Introduction

Black Friday is one of the largest annual retail shopping events, during which a retail chain experiences a sharp spike in customer transactions. This project analyzes a transactional dataset of purchases made by customers of a retail store during a Black Friday sale, with the objective of understanding customer purchase behavior across demographic and product-related attributes.

The target variable for this analysis is Purchase, which represents the amount (in the store's currency) spent by a customer on a given transaction. The goal of this report is to summarize the data preprocessing steps performed and present the findings of Univariate and Bivariate Exploratory Data Analysis (EDA) conducted with respect to the Purchase amount.

2. Objective

- Understand the structure, quality, and completeness of the dataset.
- Study the distribution of the target variable, Purchase, and detect outliers.
- Clean and preprocess the data — handle missing values, encode categorical variables, and rename fields for clarity.
- Perform Univariate Analysis to understand the distribution of individual features.
- Perform Bivariate Analysis to study the relationship between each feature and Purchase.
- Visualize key patterns using appropriate charts (histograms, boxplots, bar charts, pie charts, heatmaps).
- Summarize insights that could support business decisions such as targeted marketing and inventory planning.

3. Dataset Description

The dataset contains 550,068 transaction records across 12 columns, covering 5,891 unique customers and 3,631 unique products. Each row represents a single product purchased by a customer during the Black Friday sale.

Column	Description	Type
User_ID	Unique identifier for each customer	Identifier
Product_ID	Unique identifier for each product	Identifier
Gender	Gender of the customer (M / F)	Categorical
Age	Age group of the customer (binned range)	Categorical (Ordinal)
Occupation	Masked occupation code of the customer	Categorical (Numeric)
City_Category	Category of the city the customer resides in (A/B/C)	Categorical
Stay_In_Current_City_Years	Number of years the customer has stayed in the current city	Categorical (Ordinal)
Marital_Status	Marital status of the customer (0 = Single, 1 = Married)	Categorical (Binary)
Product_Category_1	Masked primary product category	Categorical (Numeric)
Product_Category_2	Masked secondary product category (may be absent)	Categorical (Numeric)
Product_Category_3	Masked tertiary product category (may be absent)	Categorical (Numeric)
Purchase	Purchase amount of the transaction — Target Variable	Numeric (Continuous)

Note: Occupation, Product_Category_1, Product_Category_2, and Product_Category_3 are described in the source data as "masked" — meaning their original categorical labels were already converted into numeric codes by the data provider before release.

4. Data Preprocessing

4.1 Basic Statistics

The dataset was inspected using descriptive statistics. The Purchase column ranges from a minimum of 12 to a maximum of 23,961, with a mean of approximately 9,264 and a standard deviation of approximately 5,023, indicating a moderately wide spread in transaction values.

4.2 Missing Value Check

Two columns contained missing values, both product sub-category fields:

Column	Missing Count	Missing %
Product_Category_2	173,638	31.57%
Product_Category_3	383,247	69.67%

These missing values do not indicate bad data — they occur because not every product has a second or third sub-category. Missing values were treated by filling with 0, which represents "no sub-category applicable", and the columns were then cast to integer type.

4.3 Unique Value Check

Column	Unique Values
User_ID	5,891
Product_ID	3,631
Gender	2 (M, F)
Age	7 groups
Occupation	21 codes
City_Category	3 (A, B, C)
Stay_In_Current_City_Years	5 (0, 1, 2, 3, 4+)
Marital_Status	2 (0, 1)
Product_Category_1	20 categories
Product_Category_2	17 categories
Product_Category_3	15 categories

4.4 Cleaning & Transformation Steps

- Dropped unnecessary identifier fields: User_ID and Product_ID (not informative for pattern analysis).
- Filled missing values in Product_Category_2 and Product_Category_3 with 0.
- Renamed columns for clarity — e.g. Stay_In_Current_City_Years → Stay_Years, Product_Category_1/2/3 → Product_Cat_1/2/3.
- Mapped Gender to integers using a map function: F → 0, M → 1.
- Mapped City_Category to integers: A → 0, B → 1, C → 2.
- Cleaned Stay_In_Current_City_Years by converting the '4+' label to 4 and casting to integer.
- Mapped the Age range variable into ordinal integers, e.g. 0-17 → 0, 18-25 → 1, 26-35 → 2, 36-45 → 3, 46-50 → 4, 51-55 → 5, 55+ → 6.

After these steps, the cleaned dataset had zero missing values and all columns in a numeric, model-ready format.

5. Univariate Analysis

5.1 Purchase Distribution (Target Variable)

The Purchase column was visualized using a histogram with a KDE overlay and a boxplot. The distribution is right-skewed with multiple local peaks, suggesting the presence of distinct pricing clusters (e.g. low-value accessories vs. higher-value electronics). The bulk of transactions fall between roughly 5,800 and 12,000.

Statistic	Value
Count	550,068
Mean	9,263.97
Std. Deviation	5,023.07
Minimum	12
25th Percentile (Q1)	5,823
Median (50%)	8,047
75th Percentile (Q3)	12,054
Maximum	23,961

5.2 Outlier Detection (IQR Method)

Using the Interquartile Range (IQR) method ($Q1 - 1.5 \times IQR$ to $Q3 + 1.5 \times IQR$), the calculated bounds were approximately -3,523.5 to 21,400.5. Only 2,677 transactions (about 0.49% of the dataset) fell above the upper bound as high-value outliers. Given the very small proportion, these were treated as genuine high-value purchases rather than data errors, and were retained in the dataset.

5.3 Distribution of Individual Categorical Features

- Gender: The dataset is heavily skewed toward male customers — 414,259 records (75.3%) are male vs. 135,809 (24.7%) female.
- Age: The 26-35 age group dominates with 219,587 transactions (39.9%), followed by 36-45 (110,013) and 18-25 (99,660). The youngest (0-17) and oldest (55+) groups are the smallest.
- City Category: City B has the most transactions (231,173), followed by City C (171,175) and City A (147,720).
- Marital Status: Unmarried customers (324,731) outnumber married customers (225,337).
- Stay in Current City: Customers who have stayed 1 year are the most common (193,821), followed by 2 years (101,838) and 3 years (95,285).
- Occupation: 21 distinct occupation codes are present, with counts varying considerably — some occupations are far more represented than others.
- Product_Category_1: Category 5 (150,933), Category 1 (140,378), and Category 8 (113,925) account for the large majority of all purchases.

6. Bivariate Analysis (with respect to Purchase)

6.1 Gender vs Purchase

Although male customers make far more transactions overall, the average purchase amount is also higher for males (₹9,437.53) than females (₹8,734.57) — a difference of roughly ₹700 per transaction. This suggests males not only shop more frequently on Black Friday but also make slightly higher-value purchases on average.

6.2 Age vs Purchase

Average purchase amount is relatively stable across age groups, ranging narrowly from about ₹8,933 (0-17 group) to ₹9,535 (51-55 group). The 51-55 age group recorded the highest average purchase, closely followed by 36-45 and 55+, while the youngest group (0-17) spent the least on average — though the differences across groups are modest.

6.3 Marital Status vs Purchase

Average purchase is nearly identical between single (₹9,265.91) and married (₹9,261.17) customers, indicating marital status has little to no effect on how much a customer spends per transaction.

6.4 Occupation vs Purchase

Average purchase varies moderately by occupation. Occupation codes 17 (₹9,821.48), 12 (₹9,796.64), and 15 (₹9,778.89) recorded the highest average purchase values, suggesting certain occupation segments may be more inclined toward higher-value purchases.

6.5 Purchase by City Category

City C customers spend the most on average (₹9,719.92) despite having fewer transactions than City B, followed by City B (₹9,151.30) and City A (₹8,911.94, the lowest). This indicates that while City B drives the highest transaction volume, City C drives higher average order value.

6.6 Stay in Current City (Years) vs Purchase

Average purchase shows only minor variation across the number of years a customer has stayed in their current city, suggesting this attribute is a weaker predictor of spend compared to product category or gender.

6.7 Product_Category_1 vs Purchase

Product Category 1 shows the strongest relationship with Purchase among all features analyzed. While categories 1, 5, and 8 have the highest transaction volume, they are not the highest-value categories — Category 10 has by far the highest average purchase (₹19,675.57), followed by Category 7 (₹16,365.69), Category 6 (₹15,838.48), and Category 9 (₹15,537.38). This indicates a clear split between high-volume/low-value categories and low-volume/high-value categories.

6.8 Product_Category_2 and Product_Category_3 vs Purchase

Similar to Product_Category_1, average purchase varies noticeably across the sub-category codes in Product_Category_2 and Product_Category_3, reinforcing that the specific product category — more than any customer demographic — is the strongest driver of purchase amount.

7. Data Visualization Summary

The following charts were produced as part of this analysis (see accompanying Jupyter/Colab notebook for the rendered visuals):

- Histogram + boxplot of Purchase distribution
- Bar chart of missing value percentage by column
- Count plots for Gender, Age, City Category, Marital Status, Stay Years, and Occupation
- Boxplots and average-purchase bar charts: Age vs Purchase, Gender vs Purchase, Marital Status vs Purchase
- Bar chart: Occupation vs average Purchase
- Boxplot and bar chart: City Category vs Purchase
- Bar charts: Product_Category_1, Product_Category_2, and Product_Category_3 vs average Purchase
- Pie charts: City Category distribution and Gender distribution
- Correlation heatmap across all encoded numeric features

- Multi-variable bar charts: Age $\times$ Gender vs Purchase, and Age $\times$ City Category vs Purchase
- Top 10 most frequently purchased products (by transaction count)

8. Key Insights

- The customer base is predominantly male (75.3%) and young adult (26-45 age range accounts for over half of all transactions).
- Male customers spend more per transaction on average than female customers (~₹700 difference).
- Product category is the strongest driver of purchase value — a small set of categories (10, 7, 6, 9) command significantly higher average purchase amounts than the high-volume categories (1, 5, 8).
- City B generates the most transactions, but City C customers spend more per transaction on average — a useful distinction for regional marketing strategy.
- Marital status and years of stay in the current city show minimal influence on purchase amount.
- Outliers in Purchase are minimal (0.49% of transactions) and represent genuine high-value purchases rather than data entry errors.
- Roughly 32% of transactions have no second product category and about 70% have no third product category, reflecting that most products are classified under only one or two category levels.

9. Conclusion

This exploratory analysis of the Black Friday Sales dataset shows that purchase amount is influenced far more strongly by the product being purchased (Product_Category_1/2/3) than by customer demographics such as age, marital status, or years of residence. Gender and city category show moderate influence, with male customers and City C customers associated with higher average spend. These findings can help a retailer prioritize high-value product categories in promotions, tailor city-specific marketing strategies, and better forecast revenue during peak sales events.

10. Future Scope

- Build a regression model (e.g. Linear Regression, Random Forest, XGBoost) to predict Purchase amount using the cleaned/encoded features.
- Perform customer segmentation (clustering) based on purchase behavior for targeted marketing.
- Analyze product-level associations (market basket analysis) to identify frequently co-purchased categories.
- Extend the analysis with time-based patterns if transaction timestamps become available.

11. Tools & Technologies Used

Tool / Library	Purpose
Python	Core programming language
Pandas	Data manipulation and preprocessing
NumPy	Numerical computations
Matplotlib / Seaborn	Data visualization
Google Colab	Notebook execution environment